DO NOW, AO1 REVISION

Sound waves and waves for detection and exploration

Waves
Paper 2, Separate Physics only
21 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 6 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**A sound wave passes into a different medium. Which stays the same: speed, frequency or wavelength?**	Frequency	☐
2	**State what happens to the speed of sound when it passes from air into a solid.**	It increases.	☐
3	**State what happens to the loudness of a sound passing from air into a solid, and why.**	It decreases: some sound is reflected or absorbed at the boundary.	☐
4	**Name the part of the human ear that detects sound waves.**	The eardrum	☐
5 ★	**State the range of human hearing.**	20 Hz to 20 kHz	☐
6 ★	**Name sound with a frequency above the range of human hearing.**	Ultrasound	☐
7	**What is echolocation?**	Using reflected sound (echoes) to locate an object.	☐
8	**Give two uses of ultrasound imaging.**	Medical scans (e.g. of an unborn baby) and finding flaws in materials	☐
9	**What are seismic waves?**	Waves produced by earthquakes.	☐
10	**Name the two types of seismic waves.**	P-waves and S-waves	☐
11 ★	**What type of wave are P-waves?**	Longitudinal	☐
12	**Name the states of matter that P-waves travel through.**	Solids and liquids	☐

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	QUESTION	ANSWER	✓
13 ★	**What type of wave are S-waves?**	Transverse	☐
14	**Name the state of matter that S-waves travel through.**	Solids only	☐
15	**Describe how a sound wave travels through a solid.**	The particles of the solid vibrate and pass the vibrations on.	☐
16	**Explain why human hearing is limited to a range of frequencies.**	Sound makes the eardrum vibrate only over a limited range of frequencies.	☐
17	**State what happens to ultrasound at a boundary between two media.**	It is partially reflected.	☐
18 ★	**Describe how ultrasound is used to find the distance to a boundary.**	Time the reflection: distance = speed × time ÷ 2.	☐
19	**State what echo sounding is used for.**	Detecting objects in deep water and measuring water depth.	☐
20	**State what seismic waves have provided evidence for.**	The structure and size of the Earth's core.	☐
21 ★	**Explain how S-waves provide evidence that part of the Earth's core is liquid.**	S-waves cannot travel through liquid and are not detected on the far side of the Earth.	☐

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